



VP AIR TEST NEET (2025-26)

PART TEST - 03

DROPPER

DURATION: 180 Minutes

Date: 28-09-2025

M. MARKS: 720

Topics Covered

Physics:	Laws of Motion, Work, Energy and Power, Centre of Mass & System of Particles
Chemistry:	Chemical Bonding and Molecular Structure, Solutions
Botany:	Principles of Inheritance and Variation
Zoology:	Breathing and Exchange of Gases, Body Fluids and Circulation

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

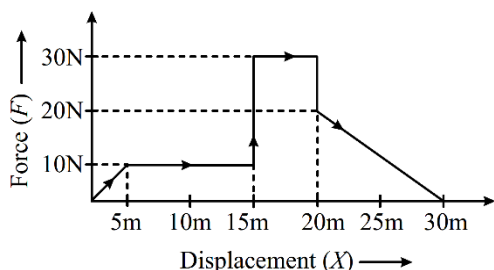
Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. When a train stops suddenly, passengers in the running train feel an instant jerk in the forward direction because
- (1) The back of seat suddenly pushes the passengers forward
 - (2) Inertia of rest stops the train and takes the body forward
 - (3) Upper part of the body continues to be in the state of motion whereas the lower part of the body in contact with seat comes to rest
 - (4) Data insufficient to conclude

2. Given below is a graph between a variable force (F) (along y -axis) and the displacement (X) (along x -axis) of a particle moving in one dimension. The work done by the force in the displacement interval between 0 m and 30 m is:



- (1) 275 J
 - (2) 375 J
 - (3) 400 J
 - (4) 300 J
3. A body of mass m falls from height h on ground. If e be the coefficient of restitution of collision between the body and ground, then the distance travelled by body before it comes to rest is:
- (1) $h \left(\frac{1+e^2}{1-e^2} \right)$
 - (2) $h \left(\frac{1-e^2}{1+e^2} \right)$
 - (3) $\frac{2eh}{1+e^2}$
 - (4) $\frac{2eh}{1-e^2}$
4. A force of magnitude F_1 acts on a particle so as to accelerate it from rest to a velocity v . The force F_1 is then replaced by another force of magnitude F_2 which decelerates it to rest from v . Then,
- (1) F_1 must be equal to F_2
 - (2) F_1 may be equal to F_2
 - (3) F_1 must be unequal to F_2
 - (4) Data insufficient

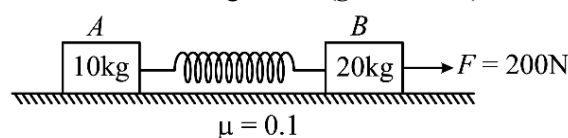
5. A force $\vec{F} = (3\hat{i} + c\hat{j} + 2\hat{k})$ N acting on a particle causes a displacement $\vec{s} = (-4\hat{i} + 2\hat{j} + 3\hat{k})$ m. If the work done by force $\vec{F}$ is 6 J, then the value of ' c ' is:
- (1) 0
 - (2) 1
 - (3) 6
 - (4) 12

6. Match List-I with List-II: (work done in complete motion)

List-I (Force-displacement graph)		List-II (Work done)	
(A)		(I)	100 J
(B)		(II)	13.5 J
(C)		(III)	15 J
(D)		(IV)	7.5 J

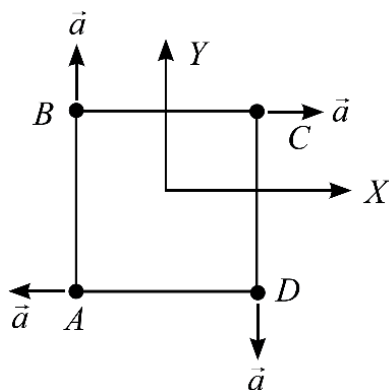
Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
 - (2) A-I, B-III, C-IV, D-II
 - (3) A-II, B-I, C-IV, D-III
 - (4) A-III, B-IV, C-I, D-II
7. Two blocks A and B attached to each other by a massless spring, are kept on a rough horizontal surface ($\mu = 0.1$) and pulled by a force $F = 200$ N as shown in figure. If at some instant the 10 kg mass has acceleration of 12 m/s^2 , what is the acceleration of 20 kg mass? ($g = 10 \text{ m/s}^2$)

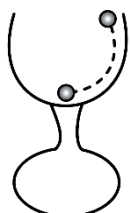


- (1) 2.5 m/s^2
- (2) 4.0 m/s^2
- (3) 3.6 m/s^2
- (4) 1.2 m/s^2

8. Four particles A, B, C and D with masses $m_A = m$, $m_B = 2m$, $m_C = 3m$ and $m_D = 4m$ are at the corners of a square. They have accelerations of equal magnitude with directions as shown. The acceleration of the centre of mass of the particles is:



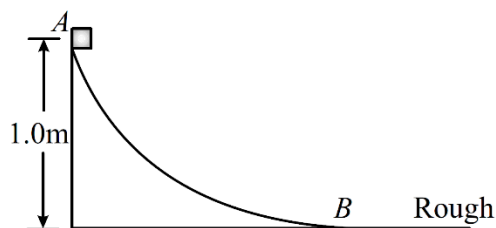
- (1) $\frac{a}{5}(\hat{i} - \hat{j})$
 - (2) Zero
 - (3) $\frac{a}{5}(\hat{i} + \hat{j})$
 - (4) $a(\hat{i} + \hat{j})$
9. A 2g ball is released from the edge of a smooth hemispherical glass whose radius is 20 cm as shown. How much work is done on the ball by the gravitational force during the ball's motion to the bottom of the glass?



- (1) 1.96 mJ
 - (2) 3.92 mJ
 - (3) 4.90 mJ
 - (4) 5.88 mJ
10. A motor cyclist moving with a velocity of 72 km/hour on a flat road takes a turn on the road at a point where the radius of curvature of the road is 20 meters. The acceleration due to gravity is 10 m/sec². In order to avoid skidding, he must not bend with respect to the vertical by an angle greater than:

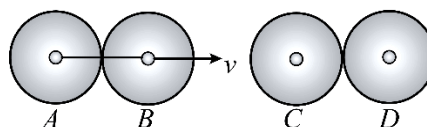
- (1) $\tan^{-1} 6$
- (2) $\tan^{-1} 2$
- (3) $\tan^{-1} 25.92$
- (4) $\tan^{-1} 4$

11. A block weighing 10 N travels down a smooth curved track AB joined to a rough horizontal surface (figure). The rough surface has a friction coefficient of 0.20 with the block. If the block starts slipping on the track from a point 1.0 m above the horizontal surface, the distance it will move on the rough surface is:



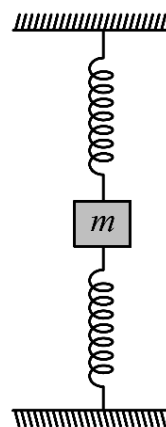
- (1) 5.0 m
- (2) 10.0 m
- (3) 15.0 m
- (4) 20.0 m

12. Two identical smooth spheres A and B are moving with same velocity v and collide with similar spheres C and D as shown, then after collision (Consider one dimensional collision and all collision are elastic)



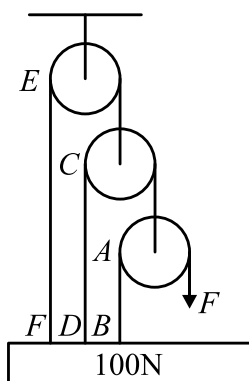
- (1) D will move with speed twice of that of C
- (2) C and D will move with same velocity v
- (3) C will stop and D will move with velocity v
- (4) All spheres A, B, C & D will move with velocity $v/2$

13. Figure shows a block tied between two massless springs in equilibrium. If upper spring is cut then the acceleration of the block just after cut is 6 m/s² downwards. Now, if instead of upper spring, lower spring is cut then the magnitude of acceleration of the block just after the cut will be: (Take $g = 10$ m/s²)



- (1) 16 m/s²
- (2) 4 m/s²
- (3) 8 m/s²
- (4) 15 m/s²

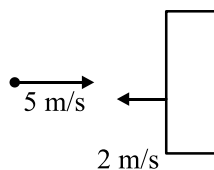
14. In the arrangement shown below force F is just sufficient to keep in equilibrium the 100 N block, T_1 , T_2 and T_3 are tension, in strings AB , CD and EF and T_4 is total sum of all tensions T_1 , T_2 and T_3 on block 100 N.



Match List-I with List-II

List-I		List-II	
(A)	T_1	(I)	$\frac{400}{7}$ N
(B)	T_2	(II)	$\frac{100}{7}$ N
(C)	T_3	(III)	$\frac{200}{7}$ N
(D)	T_4	(IV)	100 N

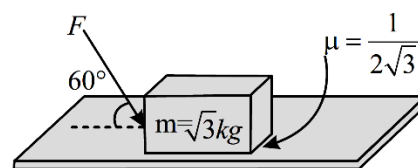
- (1) A-II, B-III, C-I, D-IV
 (2) A-I, B-III, C-II, D-IV
 (3) A-I, B-II, C-IV, D-III
 (4) A-I, B-IV, C-II, D-III
15. A ball going with speed 5 m/s collides with a heavy wall moving towards the ball at 2 m/s. If coefficient of restitution is $e = \frac{1}{2}$, then what is the speed of the ball after collision with the wall.



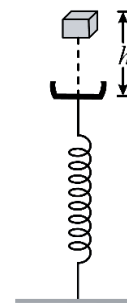
- (1) 4.5 m/s
 (2) 2.0 m/s
 (3) 5.5 m/s
 (4) 4.0 m/s
16. A body of mass 4 kg experiences two forces $\vec{F}_1 = (5\hat{i} + 8\hat{j} + 7\hat{k})$ N and $\vec{F}_2 = (3\hat{i} - 4\hat{j} - 3\hat{k})$ N. The acceleration acting on the body is:
- (1) $-2\hat{i} - \hat{j} - \hat{k}$
 (2) $4\hat{i} + 2\hat{j} + 2\hat{k}$
 (3) $2\hat{i} + \hat{j} + \hat{k}$
 (4) $2\hat{i} + 3\hat{j} + 3\hat{k}$

17. Given below are two statements:
Statement-I: An elevator can go up or down with a uniform speed when its weight is balanced with the tension of its cable.
Statement-II: Force exerted by the floor of an elevator on the foot of a person standing on it is more than his/her weight when the elevator goes down with increasing speed.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both Statement-I and Statement-II are false.
 (2) Statement-I is true but Statement-II is false.
 (3) Both Statement-I and Statement-II are true.
 (4) Statement-I is false but Statement-II is true.

18. A ball of mass m moves with speed v and it strikes normally with a wall and reflected back normally with same speed. If its time of contact with wall is t then find magnitude of force exerted by ball on wall:
- (1) $\frac{2mv}{t}$ (2) $\frac{mv}{t}$
 (3) $mv t$ (4) $\frac{mv}{2t}$
19. What is the maximum value of the force F such that the block shown in the arrangement, does not move? ($g = 10 \text{ m/s}^2$)



- (1) 20 N (2) 10 N
 (3) 12 N (4) 15 N
20. A block of mass m initially at rest is dropped from a height h on to a relaxed spring of force constant k as shown. The maximum compression in the spring is x then: (g is acceleration due to gravity)

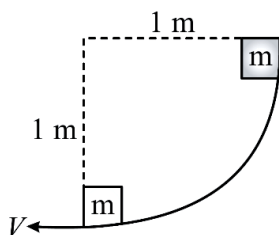


- (1) $mgh = \frac{1}{2} kx^2$
 (2) $mg(h+x) = \frac{1}{2} kx^2$
 (3) $mgh = \frac{1}{2} k(x+h)^2$
 (4) $mg(h+x) = \frac{1}{2} k(x+h)^2$

21. A body of mass m accelerates uniformly from rest to a speed v_0 in time t_0 . The work done on the body till any time t is:

(1) $\frac{1}{2}mv_0^2\left(\frac{t^2}{t_0^2}\right)$ (2) $\frac{1}{2}mv_0^2\left(\frac{t_0}{t}\right)$
 (3) $mv_0^2\left(\frac{t}{t_0}\right)$ (4) $mv_0^2\left(\frac{t}{t_0}\right)^3$

22. A block of mass 1 kg slides down a curved track which forms one quadrant of a circle of radius 1 m as shown in figure. The speed of block at the bottom of the track is $v = 2 \text{ m s}^{-1}$. The work done by the force of friction is: ($g = 10 \text{ m/s}^2$)



- (1) +4 J (2) -4 J
 (3) -8 J (4) +8 J

23. A smooth inclined plane of length L having inclination θ with the horizontal is inside a lift which is moving down with a retardation a . The time taken by a body to slide down the inclined plane from rest will be:

(1) $\sqrt{\frac{2L}{(g+a)\sin\theta}}$ (2) $\sqrt{\frac{2L}{(g-a)\sin\theta}}$
 (3) $\sqrt{\frac{2L}{a\sin\theta}}$ (4) $\sqrt{\frac{2L}{g\sin\theta}}$

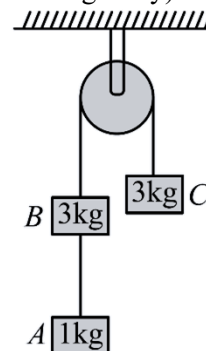
24. A small bucket containing water is rotated in a vertical circle of radius R by means of a rope. V is velocity of bucket at highest point, then water will not fall down if: (g is acceleration due to gravity)

(1) $V \leq \sqrt{gR}$ (2) $V \leq \sqrt{\frac{gR}{2}}$
 (3) $V \geq \sqrt{gR}$ (4) $V = \sqrt{\frac{gR}{2}}$

25. A particle moves with a velocity $\vec{v} = (5\hat{i} - 3\hat{j} + 6\hat{k}) \text{ m/s}$ under the influence of a constant force $\vec{F} = (10\hat{i} + 10\hat{j} + 20\hat{k}) \text{ N}$. The instantaneous power applied to the particle is:

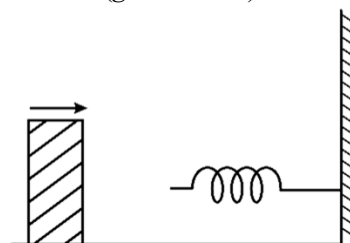
- (1) 200 J/s (2) 40 J/s
 (3) 140 J/s (4) 170 J/s

26. In the system shown in the figure, the acceleration (in m/s^2) of the 1 kg mass and the tension (in N) in the string connecting between A and B is: (g is acceleration due to gravity)



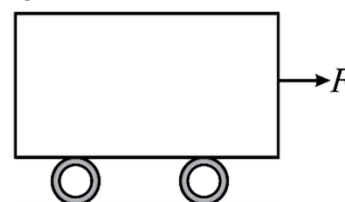
- (1) $\frac{g}{4}$ downwards, $\frac{8g}{7}$
 (2) $\frac{g}{4}$ upwards, $\frac{8}{7}$
 (3) $\frac{g}{7}$ downwards, $\frac{6}{7}g$
 (4) $\frac{g}{2}$ upwards, g

27. A 1.0 kg block collides with a horizontal weightless spring of force constant 2.75 Nm^{-1} as shown in figure. The block compresses the spring 4.0 m from the rest position. If the coefficient of kinetic friction between the block and horizontal surface is 0.25, the speed of the block at the instant of collision is: ($g = 10 \text{ m/s}^2$)



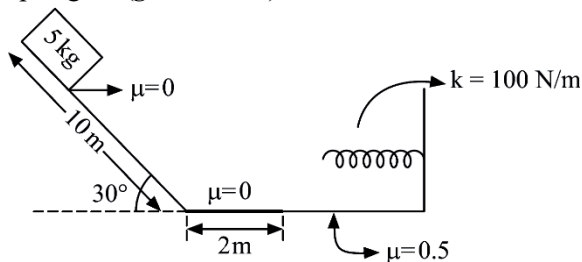
- (1) 0.4 m s^{-1} (2) 4 m s^{-1}
 (3) 0.8 m s^{-1} (4) 8 m s^{-1}

28. A car of mass m is accelerating on a level smooth road under the action of a single force F . The power delivered to the car is constant and equal to P . If the velocity of the car at an instant is v , then after travelling how much distance it becomes double?

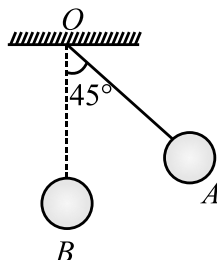


- (1) $\frac{7mv^3}{3P}$ (2) $\frac{4mv^3}{3P}$
 (3) $\frac{mv^3}{P}$ (4) $\frac{18mv^3}{7P}$

29. A block is released from the top of smooth inclined plane as shown in the figure above. The maximum compression in the spring when the block hits the spring is: ($g = 10 \text{ m/s}^2$)



- (1) $\sqrt{6} \text{ m}$ (2) 2 m
 (3) 1 m (4) $\sqrt{5} \text{ m}$
30. The bob A of a simple pendulum is released when the string makes an angle of 45° with the vertical. It hits another bob B of the same material and same mass connected to string as shown. If the collision is elastic, then after collision



- (1) Both A and B rise to the same height
 (2) Both A and B come to rest at B
 (3) Both A and B move with the same velocity of A
 (4) A comes to rest and B moves with the velocity of A

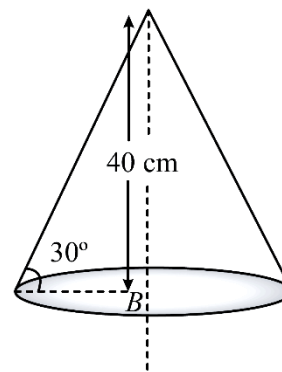
31. Potential energy as a function of r is given by,

$$U = \frac{A}{r^{10}} - \frac{B}{r^5},$$

where r is the interatomic distance, A and B are positive constants. At equilibrium the distance between the two atoms will be:

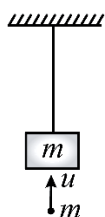
- (1) $\left(\frac{A}{B}\right)^{\frac{1}{5}}$ (2) $\left(\frac{B}{A}\right)^{\frac{1}{5}}$
 (3) $\left(\frac{2A}{B}\right)^{\frac{1}{5}}$ (4) $\left(\frac{B}{2A}\right)^{\frac{1}{5}}$
32. A simple pendulum of length 1 m has a wooden bob of mass 1 kg . It is struck by a bullet of mass 10^{-2} kg moving with a speed of $2 \times 10^2 \text{ m/s}$. The bullet gets embedded into the bob. The height to which the bob rises before swinging back is: (use $g = 10 \text{ m/s}^2$)
- (1) 0.30 m (2) 0.20 m
 (3) 0.35 m (4) 0.40 m

33. A uniform solid cone of height 40 cm is shown in figure. The distance of centre of mass of the cone from point B (centre of the base) is:

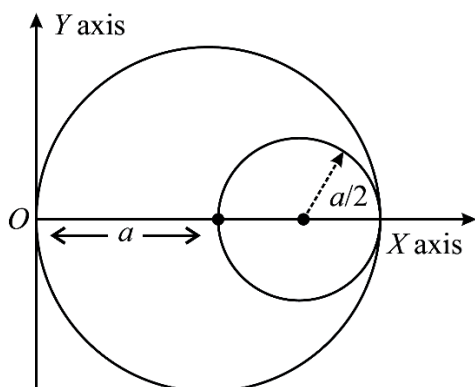


- (1) 20 cm (2) $\frac{10}{3} \text{ cm}$
 (3) $\frac{20}{3} \text{ cm}$ (4) 10 cm
34. Two objects of masses 200 gm and 500 gm possess velocities $10\hat{i} \text{ m/s}$ and $(3\hat{i} + 5\hat{j}) \text{ m/s}$ respectively. The velocity of their centre of mass in m/s is:
- (1) $5\hat{i} - 25\hat{j}$ (2) $\frac{5}{7}\hat{i} - 25\hat{j}$
 (3) $5\hat{i} + \frac{25}{7}\hat{j}$ (4) $25\hat{i} - \frac{5}{7}\hat{j}$
35. A fireman wants to slide down a rope. The rope can bear a tension of $\frac{3}{4}$ th of the weight of the man. With what minimum acceleration should the fireman slide down:
- (1) $\frac{g}{3}$ (2) $\frac{g}{6}$
 (3) $\frac{g}{4}$ (4) $\frac{g}{2}$
36. The ratio of powers of two motors is $\frac{3\sqrt{x}}{\sqrt{x}+1}$, that are capable of raising 300 kg water in 5 minutes and 50 kg water in 2 minutes respectively from a well 100 m deep. The value of x will be:
- (1) 2 (2) 4
 (3) 2.4 (4) 16
37. A particle of mass $3m$ is projected from the ground at some angle with horizontal. The horizontal range is R . At the highest point of its path it breaks into two pieces m and $2m$. The smaller mass comes to rest and larger mass finally falls at a distance x from the point of projection, where x is equal to:
- (1) $\frac{3R}{4}$ (2) $\frac{3R}{2}$
 (3) $\frac{5R}{4}$ (4) $3R$

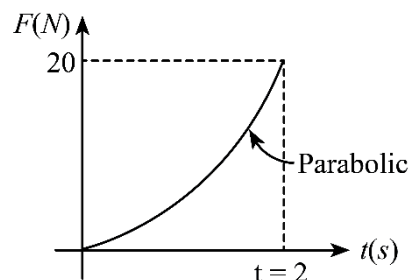
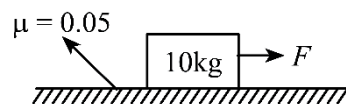
38. A bullet of mass m moving vertically upwards with a velocity ' u ' hits the hanging block of mass ' m ' and gets embedded in it. As shown in the figure the height through which block rises after the collision (assume sufficient space above block) is: (g is acceleration due to gravity)



- (1) $\frac{u^2}{2g}$ (2) $\frac{u^2}{g}$
 (3) $\frac{u^2}{8g}$ (4) $\frac{u^2}{4g}$
39. A circular hole of radius $\left(\frac{a}{2}\right)$ is cut of a circular disc of radius ' a ' as shown in figure. The centre of mass of the remaining circular portion with respect to point ' O ' will be:



- (1) $\frac{5}{6}a$ (2) $\frac{1}{6}a$
 (3) $\frac{10}{11}a$ (4) $\frac{2}{3}a$
40. A horizontal force of 49 N is just able to move a block of wood having mass 10 kg on a rough horizontal surface. The angle of friction is: (Take $g = 9.8 \text{ m/s}^2$)
 (1) $\tan^{-1}(0.5)$ (2) $\tan^{-1}(2)$
 (3) $\sin^{-1}(0.5)$ (4) 45°
41. A force F is applied to the initially stationary cart. The variation of force with time is shown in figure such that $\frac{dF}{dt} = 0$ at $t = 0$. Find the speed (m/s) of cart at $t = 2 \text{ s}$. ($g = 10 \text{ m/s}^2$)



- (1) 4 (2) 0.67
 (3) 3.75 (4) 2
42. **Statement I:** The centre of mass of an object always lies at the geometric centre of the object.
Statement II: The centre of mass always coincides with the centre of gravity.
 (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.
43. Three solids of masses m_1, m_2 and m_3 are connected with weightless string in succession and are placed on a frictionless table. If the mass m_3 is dragged with a pulling force T , the tension in the string between m_2 and m_3 is:
 (1) $\frac{m_2}{m_1 + m_2 + m_3}T$ (2) $\frac{m_3}{m_1 + m_2 + m_3}T$
 (3) $\frac{m_1 + m_2}{m_1 + m_2 + m_3}T$ (4) $\frac{m_2 + m_3}{m_1 + m_2 + m_3}T$
44. A small block is lying at rest on a stationary truck. The contact surface between the block and truck is rough with coefficient of static and kinetic friction, 0.3 and 0.2 respectively. What is the maximum acceleration with which the truck can start moving such that the block does not slip with respect to the truck? ($g = 9.8 \text{ m/s}^2$)
 (1) 3.00 m/s^2 (2) 2.94 m/s^2
 (3) 2.00 m/s^2 (4) 1.96 m/s^2
45. The upper half of an inclined plane of inclination θ is perfectly smooth while the lower half is rough. A body starting from the rest at top comes back to rest at the bottom if the coefficient of friction for the lower half is given by:
 (1) $\mu = \sin \theta$ (2) $\mu = \cot \theta$
 (3) $\mu = 2 \cos \theta$ (4) $\mu = 2 \tan \theta$

46. A 1.5 molal solution of KOH (aq) has a density 0.8 g cm^{-3} . The molarity of the solution is:
 (Given: Atomic masses: K = 39 u, O = 16 u, H = 1 u)
- (1) 1.86 M
 - (2) 1.11 M
 - (3) 2.34 M
 - (4) 0.62 M

47. Lattice enthalpy of NaCl means the energy required to separate one mole of solid NaCl into:
- (1) one mole of $\text{Na}^+(\text{g})$ and one mole of $\text{Cl}^-(\text{g})$
 - (2) one mole of $\text{Na}(\text{s})$ and half mole of $\text{Cl}_2(\text{g})$
 - (3) one mole of $\text{Na}(\text{g})$ and half mole of $\text{Cl}_2(\text{g})$
 - (4) one mole of $\text{Na}(\text{s})$ and one mole of $\text{Cl}(\text{g})$

48. Given below are two statements:

Statement I: Vapour pressure of a liquid increases linearly with increasing temperature.

Statement II: Vapour pressure is a colligative property.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are incorrect.

49. Match List-I with List-II.

List-I (Classification of molecules based on octet rule)		List-II (Example)	
(A)	Molecule obeying octet rule	(I)	NO_2
(B)	Molecule with incomplete octet and odd electron	(II)	BeH_2
(C)	Molecule with incomplete octet and no odd electron	(III)	SF_6
(D)	Molecule with expanded octet	(IV)	PCl_3

Choose the **correct** answer from the options given below:

- (1) A-III; B-IV; C-I; D-II
- (2) A-III; B-IV; C-II; D-I
- (3) A-IV; B-I; C-II; D-III
- (4) A-IV; B-III; C-I; D-II

50. A solution is prepared by adding 4 g of solute (X) in two mole of water. Mass percentage of (X) in the solution is:

- (1) 5%
- (2) 10%
- (3) 15%
- (4) 20%

51. Consider the following statements regarding H-bond:

- A. Magnitude of H-bonding is maximum in solid state of a compound
- B. Hydrogen bonds have strong influence on the structure and properties of the compounds
- C. Intramolecular H-bonding is present in HF

The **correct** statements is/are:

- (1) B only
- (2) A and B only
- (3) A and C only
- (4) A, B and C

52. If partial pressure of CO_2 in atmosphere is 0.8 bar and Henry's law constant for CO_2 in water is 1.67 kbar at 298K, then the amount of CO_2 dissolve in one litre water to prepare soda water is nearly:

- (1) 1.14 g
- (2) 1.98 g
- (3) 2.84 g
- (4) 0.12 g

53. **Correct** order of increasing dipole moment for the given compounds is:

- (1) $\text{NH}_3 < \text{H}_2\text{O} < \text{HF}$
- (2) $\text{H}_2\text{O} < \text{NH}_3 < \text{HF}$
- (3) $\text{HF} < \text{NH}_3 < \text{H}_2\text{O}$
- (4) $\text{NH}_3 < \text{HF} < \text{H}_2\text{O}$

54. The observed and normal molar masses of compound MX_2 are 70 and 170 respectively. The percentage degree of ionisation of ternary electrolyte MX_2 is:

- (1) 83.8%
- (2) 65.8%
- (3) 55.2%
- (4) 71.4%

55. The **correct** order of polarising power of given cations is:

- (1) $\text{Ca}^{2+} < \text{Mg}^{2+} < \text{K}^+$
- (2) $\text{Mg}^{2+} < \text{Ca}^{2+} < \text{K}^+$
- (3) $\text{K}^+ < \text{Mg}^{2+} < \text{Ca}^{2+}$
- (4) $\text{K}^+ < \text{Ca}^{2+} < \text{Mg}^{2+}$

56. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: A pressure cooker reduces cooking time for food.

Reason R: Boiling point of water involved in cooking is increased.

In the light of above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true and R is NOT the correct explanation of A.

57. Which among the following molecules has sp^3d hybridization, different bond lengths and lone pair on central atom?

- (1) XeF_4
- (2) ClF_3
- (3) BrF_5
- (4) PF_5

58. 12 g of a non-volatile solute dissolved in 90 g of water produces the relative lowering of vapour pressure of 0.2. The molecular mass of the solute is:

- (1) 24 u
- (2) 12 u
- (3) 32 u
- (4) 40 u

59. Consider the following statements for ethene molecule:

A. The H-C-H bond angle is 117.6°

B. The H-C-C bond angle is 121°

C. The C-H bond is sp^2 -s sigma bond.

The **correct** statements are:

- (1) A and B only
- (2) B and C only
- (3) A and C only
- (4) A, B and C

60. Match List-I with List-II.

List-I		List-II	
(A)	Solution of chloroform and acetone.	(I)	Ebullioscopic constant
(B)	Solution of ethanol and water	(II)	Cryoscopic constant
(C)	Molal depression constant	(III)	Minimum boiling azeotrope
(D)	Molal elevation constant	(IV)	Maximum boiling azeotrope

Choose the **correct** answer from the options given below:

- (1) A-III; B-IV; C-I; D-II
- (2) A-IV; B-III; C-II; D-I
- (3) A-IV; B-III; C-I; D-II
- (4) A-III; B-IV; C-II; D-I

61. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Positive orbital overlap is the in phase overlap.

Reason R: It results due to overlapping of orbitals having same sign and orientation in space.

In the light of above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true and R is NOT the correct explanation of A.

62. **Incorrect** statement among the following is:

- (1) The concentration of pollutants in water is often expressed in terms of ppm.
- (2) A 35% (V/V) solution of ethylene glycol is used in cars for cooling the engine.
- (3) Commercial bleaching solution contains 3.62 mass percentage of sodium chlorate in water.
- (4) Commonly used unit in medicine and pharmacy is mass by volume percentage.

63. Paramagnetic species among the following is:

- (1) B_2
- (2) N_2
- (3) F_2
- (4) Li_2

64. **Incorrect** match among the following is:

- (1) Solid solution: Solution of hydrogen in palladium.
- (2) Gaseous solution: Camphor in nitrogen gas.
- (3) Solid solution: Copper dissolved in gold.
- (4) Liquid solution: Amalgam of mercury with sodium.

65. How many compounds among the following can form KHX_2 type salt? (where, X = F, Cl, Br, I)

HF, HCl, HBr, HI

- (1) 4
- (2) 3
- (3) 2
- (4) 1

66. For a solution of bromoethane and chloroethane, how many thermodynamic parameters among the following are **correct**?

$\Delta_{mix} H = 0$, $\Delta_{mix} G < 0$, $\Delta_{mix} V = 0$, $\Delta_{mix} S > 0$.

- (1) 4
- (2) 3
- (3) 2
- (4) 1

67. Both ionic and covalent bonds are present in:
 (1) HNO_3 (2) BCl_3
 (3) C_2H_6 (4) K_2CO_3
68. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Water boils at 373.15 K.
Reason R: At 100°C the vapour pressure of water is 1 bar.
 In the light of above statements, choose the **correct** answer from the options given below:
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true and R is NOT the correct explanation of A.
69. Bond order of CO is:
 (1) 2 (2) 2.5
 (3) 3 (4) 3.5
70. Which one of the following aqueous solutions will exhibit highest boiling point?
 (1) 0.01 M urea. (2) 0.15 M glucose
 (3) 0.10 M KNO_3 (4) 0.02 M Na_2SO_4
71. Bond angle in SO_2 molecule is:
 (1) 119.5° (2) 120°
 (3) 120.5° (4) 119°
72. 0.02 M AgNO_3 solution when treated with 0.01 M NaCl solution gives white ppt of AgCl . The two solutions are separated as shown below:
- | | | |
|-----------------------------|-----|---------------------------|
| AgNO_3
solution | SPM | NaCl
solution |
|-----------------------------|-----|---------------------------|
- (where, SPM: semi permeable membrane)
 Due to osmosis:
 (1) white ppt formation is observed on side X
 (2) white ppt formation is observed on side Y
 (3) molarity of AgNO_3 solution is lowered
 (4) molarity of NaCl solution is lowered
73. Resonance is absent in:
 (1) CO_2 (2) CO_3^{2-}
 (3) NH_4^+ (4) NO_2
74. A 5% solution of substance is isotonic with 2.5% solution of urea (molar mass = 60 g mol^{-1}). The molar mass of the substance will be:
 (1) 75 g mol^{-1} (2) 30 g mol^{-1}
 (3) 120 g mol^{-1} (4) 45 g mol^{-1}

75. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: C_2 consists two pi bonds.
Reason R: It is due to presence of four electrons in two pi molecular orbitals.
 In the light of above statements, choose the **correct** answer from the options given below:
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true and R is NOT the correct explanation of A.
76. The vapour pressure of pure benzene and toluene at 27°C are 80 torr and 24 torr respectively. The mole fraction of toluene in vapour phase, in equilibrium with 1:3 molar ratio of benzene and toluene solution at the same temperature is:
 (1) 0.56
 (2) 0.38
 (3) 0.47
 (4) 0.63
77. Maximum average bond length among the following is of:
 (1) O–H (2) N–O
 (3) C–N (4) C–C
78. Least soluble gas in water at 298 K among the following is:
 (1) Ar
 (2) HCHO
 (3) CH_4
 (4) $\text{CH}_2=\text{CH}-\text{Cl}$
79. Given below are two statements:
Statement I: Formal charges help in the selection of the lowest energy structure from a number of possible Lewis structures for a given species.
Statement II: Ionic bonds will be formed more easily between elements with comparatively low ionization enthalpies and elements with comparatively high negative value of electron gain enthalpy.
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
 (1) Both Statement I and Statement II are correct.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Both Statement I and Statement II are incorrect.

80. Consider the following statements:
- The tanks used by scuba divers are filled with air diluted with helium.
 - Excess of salty food results a situation called edema.
 - 1% (mass/volume) NaCl solution is called normal saline solution.
 - The solubility of a solid in a liquid is significantly affected by temperature.
 - Naphthalene dissolve readily in water.

The **correct** statements are:

- A, D and E
 - A, B, D and E
 - B, C, and E
 - A, B and D
81. Dative bond is **not** present in:
- CO
 - NO_3^-
 - O_3
 - NO
82. A sample of drinking water contain 15 ppm (by mass) chloroform. The percentage by mass of chloroform in the sample is:
- $1.5 \times 10^{-3}\%$
 - $1.5 \times 10^{-2}\%$
 - $1.5 \times 10^{-4}\%$
 - $1.5 \times 10^{-5}\%$
83. Number of σ -bonds in $\text{C}_2(\text{CN})_4$ molecule is:
- 9
 - 8
 - 10
 - 11
84. x g of benzoic acid (molar mass = 122 g mol^{-1}) dissolved in 70 g of benzene ($K_f = 5.12 \text{ K kg mol}^{-1}$) shows a depression in freezing point equal to 1 K. If the percentage association of the acid to form dimer is 60%, then x is:
- 1.46 g
 - 3.11 g
 - 2.38 g
 - 4.26 g
85. Lattice energy is maximum for:
- MgO
 - Al_2S_3
 - CaCl_2
 - NaI

86. The vapour pressure observed by mixing two liquid A and B is 400 mm of Hg when their mole fractions are 0.8 and 0.2 respectively in the solution. The total vapour pressure becomes 450 mm of Hg if mole fractions are changed to 0.2 and 0.8 respectively for A and B. The vapour pressure of B in pure state is:
- 423.23 mm of Hg
 - 466.66 mm of Hg
 - 481.26 mm of Hg
 - 407.12 mm of Hg

87. Match List-I with List-II.

List-I (Ion)		List-II (Shape)	
(A)	I_3^-	(I)	Bent
(B)	BF_4^-	(II)	Linear
(C)	NH_2^-	(III)	Trigonal pyramidal
(D)	H_3O^+	(IV)	Tetrahedral

Choose the **correct** answer from the options given below:

- A-II; B-IV; C-I; D-III
 - A-II; B-III; C-I; D-IV
 - A-IV; B-III; C-I; D-II
 - A-III; B-IV; C-II; D-I
88. Given below are two statements:
- Statement I:** Molality is independent of temperature.
- Statement II:** Reverse osmosis is used in desalination of sea water.
- In the light of the above statements, choose the *most appropriate* answer from the options given below:
- Both Statement I and Statement II are correct.
 - Statement I is correct but Statement II is incorrect.
 - Statement I is incorrect but Statement II is correct.
 - Both Statement I and Statement II are incorrect.
89. During the conversion of O_2^- to O_2^{2-} , the incoming electron goes to which orbital?
- $\pi 2p$ orbital
 - $\pi^* 2p$ orbital
 - $\sigma 2p$ orbital
 - $\sigma^* 2p$ orbital
90. When 800 mL of 0.5 M nitric acid (molar mass = 63 g mol^{-1}) is heated in a beaker, its volume is reduced to half and 10.2 g of nitric acid is evaporated. The molarity of resulting nitric acid solution is:
- 0.59 M
 - 0.42 M
 - 0.68 M
 - 0.33 M

91. In the Mendelian dihybrid cross;
- (1) 9:3:3:1 ratio is obtained for the phenotype at F_1 .
 - (2) the alleles of the genes for two traits segregate similar to those in a monohybrid cross.
 - (3) the genes of the two traits do not assort independently.
 - (4) the progeny of the F_1 generation resembled both the parents.

92. Consider the following statements regarding why Mendel's work remained unrecognised for a long time.
- A. His work was not widely publicised as communication was not easy in those days.
 - B. His concept of genes as discrete units that did not blend was not accepted by his contemporaries.
 - C. His approach of using mathematical logic to explain biological phenomena was outdated.
 - D. He failed to provide any physical evidence for the existence of factors or their composition.
- How many of the above statements are **incorrect**?
- (1) One
 - (2) Two
 - (3) Three
 - (4) Four

93. Match List-I with List-II.

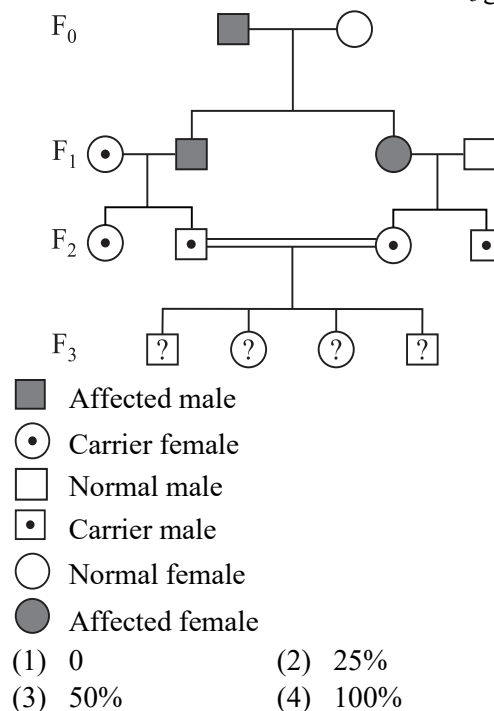
List I		List II	
(A)	Haemophilia	(I)	One extra sex chromosome
(B)	Klinefelter's Syndrome	(II)	Mutation in sex chromosome
(C)	Turner's syndrome	(III)	One sex chromosome absent
(D)	Myotonic dystrophy	(IV)	Mutation in autosome

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-IV, D-II
 - (2) A-III, B-IV, C-II, D-I
 - (3) A-II, B-I, C-III, D-IV
 - (4) A-IV, B-II, C-I, D-III
94. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
- Assertion (A):** Phenylketonuria is a genetic disorder that demonstrates the phenomenon of pleiotropy.
- Reason (R):** The mutated gene responsible for phenylketonuria affects multiple phenotypic expressions, including mental retardation and reduced pigmentation.
- In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

95. With the help of given pedigree, find out the probability for the birth of a child who is neither a carrier nor is affected with disease in F_3 generation.



96. Given below are two statements:
- Statement I:** The Law of Segregation holds true even in cases of incomplete dominance.
- Statement II:** In case of incomplete dominance, the parental alleles do not segregate during gamete formation.
- In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both Statement I and Statement II are correct.
 - (2) Both Statement I and Statement II are incorrect.
 - (3) Statement I is correct but Statement II is incorrect.
 - (4) Statement I is incorrect but Statement II is correct.
97. In pea plants, the gene for starch synthesis has two alleles, B and b. Identify the **false** statement about it.
- (1) BB homozygotes produce large starch grains.
 - (2) bb homozygotes produce wrinkled seeds.
 - (3) Starch grain size shows complete dominance and seed shape shows incomplete dominance.
 - (4) Bb heterozygote produces round seeds but starch grains of intermediate size.

98. Arrange the following steps of an artificial hybridization experiment, as performed by Mendel in pea, in the correct sequence.

- Dusting of pollen from the anther of the male parent on the stigma of the female parent.
- Collection of seeds from the resulting fruit.
- Emasculation of the female bisexual parent.
- Growing the seeds to produce the first hybrid generation.
- Choosing two true-breeding lines.

Choose the **correct** option.

- A – E – C – B – D
- E – C – A – B – D
- C – A – E – D – B
- A – B – C – E – D

99. Following Mendel's work on inheritance patterns, what became a primary focus for biologists in the subsequent century?

- The mechanism of evolution through natural selection.
- The classification of organisms based on morphological traits.
- The study of ecosystems and environmental interactions.
- Understanding the structure of the genetic material, the structural basis of genotype and phenotype conversion.

100. Identify the **correctly** matched pair.

- Alleles – Genes which code for a pair of contrasting traits
- Phenotype – The genetic constitution of an organism, e.g., Tt
- Homozygous – An organism containing two different alleles for a particular trait
- Dominant factor – An allele which expresses itself only in the homozygous state

101. Match **List-I** with **List-II**.

List I		List II	
(A)	Henking	(I)	Experimental verification of chromosomal theory of inheritance
(B)	Sutton	(II)	Studied spermatogenesis in a few insects and traced 'X body'
(C)	T.H. Morgan	(III)	Used recombination frequency to map gene positions on a chromosome
(D)	Alfred Sturtevant	(IV)	Proposed the chromosomal theory of inheritance

Choose the *most appropriate* answer from the options given below.

- A-II, B-IV, C-I, D-III
- A-II, B-I, C-IV, D-III
- A-III, B-IV, C-I, D-II
- A-I, B-II, C-III, D-IV

102. Read the following statements about Klinefelter's Syndrome.

- It is caused by the absence of one X chromosome.
- The individual has a karyotype of 47, XXY.
- Affected individuals exhibit overall masculine development along with gynaecomastia.
- Such individuals are fertile and can reproduce normally.
- It is a case of polyploidy.

In the light of above statements choose the **correct** answer from the options given below:

- B and C are correct
- A and D are correct
- A, B and C are incorrect
- B, C and E are incorrect

103. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): To determine the phenotype of a tall pea plant at F₂, Mendel performed a test cross.

Reason (R): A test cross involves crossing the test organism with a recessive parent.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- Both (A) and (R) are true and (R) is the correct explanation of (A).
- Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (A) is true but (R) is false.
- (A) is false but (R) is true.

104. A couple had 4 children. Each had a different blood group. What is the possible combination of genotypes of parents for blood group?

- ii and ii
- $I^A i$ and ii
- $I^B i$ and ii
- $I^A i$ and $I^B i$

105. Given below are two statements:

Statement I: In polygenic inheritance, traits are spread across a gradient.

Statement II: In a polygenic trait the effect of each allele is additive.

In the light of the above statements, choose the **correct** answer from the options given below:

- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.
- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.

- 106.** Match **List-I** with **List-II** w.r.t a monohybrid cross between true-breeding lines.

List I		List II	
(A)	Pink coloured flowers in dogflower at F_1	(I)	25%
(B)	Green coloured pea seed at F_1	(II)	100%
(C)	Red coloured flowers in snapdragon at F_2	(III)	75%
(D)	Yellow coloured pea seed at F_2	(IV)	0%

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-II, D-IV
 (2) A-II, B-IV, C-I, D-III
 (3) A-IV, B-I, C-II, D-III
 (4) A-I, B-II, C-III, D-IV
- 107.** The **correct** set of recessive traits in pea studied by Mendel is;
- (1) tall height, white flower and green pod.
 (2) violet flower, terminal flower position and yellow pod.
 (3) dwarf height, wrinkled seed and constricted pod.
 (4) axial flower position, green seed and inflated pod.
- 108.** Read the following statements regarding ABO blood grouping in humans.
- A. It is controlled by gene I that has three alleles.
 B. It is an example of multiple allelism, incomplete dominance and co-dominance.
 C. A person with blood group 'O' has RBCs that lack surface sugars.
 D. A total of six different genotypes and four phenotypes are possible for this character.
 E. The alleles I^A and I^B produce a slightly different form of the sugar.
- In the light of above statements choose the **correct** answer from the options given below:
- (1) A, C, D and E are correct
 (2) B, C, D and E are correct
 (3) A, B, C and D are incorrect
 (4) A, B, D and E are incorrect
- 109.** In grasshoppers;
- (1) half of the male gametes have Y chromosome.
 (2) the males have half the number of chromosomes as compared to females.
 (3) females have two sex chromosomes while males have only one, besides autosomes.
 (4) the females are heterogametic.

- 110.** Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Both thalassemia and haemophilia are disorders associated with blood.

Reason (R): Both thalassemia and haemophilia are autosomal recessive disorders.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 (3) (A) is true but (R) is false.
 (4) (A) is false but (R) is true.

- 111.** Choose the **correctly** matched pair.

- (1) Normal human : 46 autosomes
 (2) Down's syndrome : 44 autosomes
 (3) Turner's syndrome : 44 autosomes
 (4) Klinefelter's syndrome : 47 autosomes

- 112.** In an organism, the frequency of recombination between three gene pairs (y, w, m) on the same chromosome was determined.

- (i) between y and w: 1.3%
 (ii) between y and m: 37.2%
 (iii) between w and m: 35.9%

Based on this information, what is the correct linear order of the genes on the chromosome?

- (1) w – y – m
 (2) y – w – m
 (3) m – y – w
 (4) y – m – w

- 113.** Given below are two statements:

Statement I: When Morgan hybridised yellow-bodied, white-eyed females to brown-bodied, red-eyed male *Drosophila*, 37.2% of progeny were recombinants in F_2 .

Statement II: When Morgan hybridised female *Drosophila* with white eye and miniature wing with males having red-eye and normal wing, 62.8% of progeny were parental types in F_2 .

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
 (2) Both Statement I and Statement II are incorrect.
 (3) Statement I is correct but Statement II is incorrect.
 (4) Statement I is incorrect but Statement II is correct.

114. The sex chromosomes in birds are represented as;

- (1) Male:  ; Female: 
- (2) Male:  ; Female: 
- (3) Male:  ; Female: 
- (4) Male:  ; Female: 

115. Match List-I with List-II.

List I		List II	
(A)	Height in humans	(I)	Autosomal dominant Mendelian disorder
(B)	Phenylketonuria	(II)	Chromosomal disorder
(C)	Down's syndrome	(III)	Pleiotropy
(D)	Myotonic dystrophy	(IV)	Polygenic trait

Choose the *most appropriate* answer from the options given below.

- (1) A-IV, B-III, C-II, D-I
 (2) A-III, B-II, C-IV, D-I
 (3) A-II, B-I, C-IV, D-III
 (4) A-IV, B-I, C-II, D-III
116. A test cross was performed on a pea plant with round seed shape. All the offsprings had round seed shape. What can be inferred about the genotype of the test plant?
- (1) It is homozygous dominant (RR).
 (2) It is homozygous recessive (rr).
 (3) It is heterozygous (Rr).
 (4) The result is inadequate to predict the genotype.
117. Which of the following statements correctly describe the Law of Dominance proposed by Mendel?
- A. Characters are controlled by discrete units called factors.
 B. Factors always occur in pairs in an individual.
 C. In a dissimilar pair of factors, one member of the pair dominates the other.
 D. This law explains why alleles blend in a heterozygous individual.
- Choose the **correct** option.
- (1) A, B and C
 (2) A and C only
 (3) B and D only
 (4) A, C and D

118. A woman who is a carrier for haemophilia marries a normal man. Among all the progeny, what is the probability of their daughter being haemophilic?

- (1) 0% (2) 25%
 (3) 50% (4) 100%

119. In sickle cell anaemia;

- (1) alpha globin chain of haemoglobin molecule is affected.
 (2) point mutation is observed.
 (3) the wild haemoglobin molecule undergoes polymerisation under low oxygen tension.
 (4) GUG is substituted by GAG in the beta globin chain of haemoglobin molecule.

120. Given below are two statements:

Statement I: Pedigree analysis cannot be done for Turner's syndrome.

Statement II: Pedigree analysis can be done for cystic fibrosis.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
 (2) Both Statement I and Statement II are incorrect.
 (3) Statement I is correct but Statement II is incorrect.
 (4) Statement I is incorrect but Statement II is correct.

121. Read the following regarding the behaviour of genes and chromosomes.

- A. They occur in pairs in diploid organisms.
 B. They segregate during gamete formation.
 C. Only one of each pair is transmitted to a gamete.
 D. Independent gene/chromosome pairs can never segregate independently of each other.
 E. The two alleles of a gene pair are located on homologous sites on homologous chromosomes.

The **incorrect** statement is;

- (1) D (2) E
 (3) A (4) B

122. In the F₂ generation of a Mendelian dihybrid cross, the ratio of parental type to recombinant type progeny was;

- (1) 3:5 (2) 5:3
 (3) 1:16 (4) 16:1

123. Choose the **incorrect** statement.

- (1) In a Punnett square, the possible gametes are written on two sides.
 (2) A Punnett square cannot be used for a dihybrid cross.
 (3) Reginald C. Punnett was a British geneticist.
 (4) In a Punnett square, the possible gametes are usually written on the top row and left columns.

124. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In human beings, it is the genetic makeup of the sperm that determines the sex of the child.

Reason (R): Human males produce two different types of gametes, 50 per cent carrying the X-chromosome and 50 per cent carrying the Y-chromosome.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

125. Thomas Hunt Morgan selected *Drosophila melanogaster* for his genetic studies. Which of the following features contributed to its suitability?

- A. They can be grown on simple synthetic medium.
- B. They have a very long life cycle of about two months.
- C. A single mating produces a small number of progeny.
- D. Male and female flies are easily distinguishable.
- E. It has many types of hereditary variations visible with low power microscopes.

Select the **correct** combination of statements.

- (1) C, D, E (2) A, B, C
- (3) A, D, E (4) B, C, D

126. The **correct** set of characters in a wild fruit fly is;

- (1) red eye, yellow body and miniature wing.
- (2) white eye, brown body and normal wing.
- (3) white eye, brown body and miniature wing.
- (4) red eye, brown body and normal wing.

127. Choose the **correctly** matched pair.

- (1) Sahiwal : Breed of cow created by natural selection
- (2) Mendel : Performed artificial hybridisation experiments on pea from 1856-1863
- (3) *Antirrhinum* sp.: Used by Mendel for dihybrid crosses
- (4) Queen Victoria : Carrier of colour blindness

128. In the inborn error of metabolism;

- (1) single gene mutation is observed.
- (2) the gene for enzyme phenyl alanine hydroxylase is overexpressed.
- (3) phenylpyruvic acid is not synthesised.
- (4) X-chromosome is affected.

129. In honey bees, the males;

- (1) have father and sons.
- (2) cannot have grandfather and grandsons.
- (3) produce sperms by mitosis.
- (4) are called workers.

130. Match the **List-I** with **List-II**.

List I		List II	
(A)	α Thalassemia	(I)	Chromosome 11
(B)	β Thalassemia	(II)	Chromosome 16
(C)	Colour blindness	(III)	Chromosome 21
(D)	Down's syndrome	(IV)	Chromosome X

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-IV, D-II
- (2) A-I, B-III, C-II, D-IV
- (3) A-II, B-I, C-IV, D-III
- (4) A-I, B-III, C-IV, D-II

131. A cell having two pairs of chromosomes namely A and its homologous A' and B and its homologous B' undergoes meiosis. Which of the following combinations of chromosomes is usually not present in its gamete?

- (1) A-B (2) A-B'
- (3) A'-B' (4) B-B'

132. Identify the **correct** statements.

- A. Mutation is defined as change in the genetic material.
- B. Inheritable mutations can be studied by generating a pedigree of a family.
- C. Change in a subset of chromosome number is called inheritance.
- D. Analysis of karyotypes enables us to identify the number of chromosomes in an individual.
- E. Polyploidy and aneuploidy helped in understanding the mutational basis of genetic disorders.

Choose the **correct** option.

- (1) A, B, D, E (2) B, C, D, E
- (3) A, B, C, D (4) A, C, D, E

133. The approximate percentage of occurrence of colour blindness in human males and females, respectively is;

- (1) 8 and 0.4 (2) 0.4 and 8
- (3) 8 and 8 (4) 0.4 and 0.4

134. In his monohybrid cross between tall and dwarf pea plants, what did Mendel observe in the F₁ progeny?

- (1) All plants were of intermediate height.
- (2) 50% of the plants were tall and 50% were dwarf.
- (3) All plants were tall, resembling one of the parents.
- (4) All plants were dwarf, resembling the recessive parent.

135. Choose the **incorrect** statement.

- (1) The chromosomes which are different in two sexes are named as autosomes while those that were similar are named sex chromosomes.
- (2) A recessive character that was not expressed in heterozygous condition may be expressed again when it becomes homozygous.
- (3) Mendel found that the factors independently assort and combine in all permutations and combinations.
- (4) Mendel's law of independent assortment does not hold true for the genes that were located on the same chromosomes.

136. A patient's spirometry report shows a Vital Capacity (VC) of 4800 ml, a Tidal Volume (TV) of 500 ml, and an Expiratory Reserve Volume (ERV) of 1200 ml. Based on these values, what is the patient's Inspiratory Capacity (IC)?

(1) 3100 ml (2) 3600 ml
(3) 4300 ml (4) 3500 ml

137. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The process of blood coagulation is a complex cascade that can be initiated through more than one pathway following an injury.

Reason R: Coagulation is triggered by a series of linked enzymatic reactions involving factors that are always present in the plasma in their active state.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

138. Read the following statements carefully and choose the option with the **correct** statements.

A. The breakdown of simple molecules like glucose and amino acids is a catabolic process that directly utilizes O_2 to produce energy.

B. Carbon dioxide, a by-product of metabolic activities, is essential and must be retained within the cells.

C. The process of exchanging atmospheric O_2 with CO_2 produced by the cells is termed breathing.

D. The terms 'breathing' and 'respiration' can be used interchangeably in a common context but they are not same.

(1) A and B only
(2) C and D only
(3) A, C and D only
(4) A and C only

139. Given below are two statements.

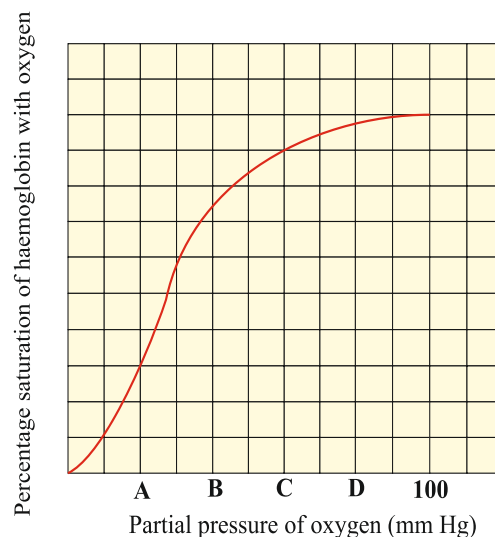
Statement I: In higher vertebrates, blood is classified as the primary tissue fluid responsible for the circulation of O_2 , nutrients, and other essential substances.

Statement II: Plasma, the fluid matrix of blood, is primarily composed of 55-70% water, with proteins making up 6-8% of its total volume.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

140. In the given diagram of the oxygen dissociation curve, under normal physiological conditions, the partial pressure of oxygen at the tissue level is represented by:



(1) A
(2) B
(3) C
(4) D

141. Match List-I with List-II.

List-I (Leucocyte)		List-II (Nucleus Shape)	
(A)	Neutrophil	(I)	Kidney-shaped
(B)	Eosinophil	(II)	S-shaped
(C)	Basophil	(III)	Multi-lobed
(D)	Monocyte	(IV)	Bilobed

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-II, D-I
(2) A-I, B-II, C-III, D-IV
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-III, C-I, D-II

142. How many of the following is/are **incorrectly** matched with respect to its function in the regulation of respiration?

- A. Respiratory Rhythm Centre: Located in the medulla, it is the primary center responsible for generating the basic inspiratory and expiratory rhythm.
- B. Pneumotaxic Centre: Located in the pons region, its signals can moderate the function of the rhythm center, primarily by reducing the duration of inspiration.
- C. Aortic and Carotid Bodies: These peripheral chemoreceptors are primarily stimulated by an increase in arterial $p\text{CO}_2$ and H^+ concentration.
- D. Central Chemoreceptors: Located adjacent to the rhythm center in the medulla, they are highly sensitive to changes in both O_2 and CO_2 concentration in the arterial blood.

- (1) One
- (2) Two
- (3) Three
- (4) Four

143. Identify the **correct** sequence for the duration of cardiac cycle events, from longest to shortest.

(1)	Auricular systole	Joint diastole	Ventricular Systole
(2)	Ventricular systole	Joint diastole	Auricular systole
(3)	Ventricular systole	Auricular systole	Joint diastole
(4)	Joint diastole	Ventricular systole	Auricular systole

144. Choose the option that **correctly** identifies the pair(s) as correct or incorrect w.r.t humans.

(A)	Capillary	(I)	Open ended
(B)	Heart	(II)	Pumping station of the body
(C)	Lymph	(III)	Helps in fat absorption
(D)	Rh antigen	(IV)	Present in Rhesus kangaroos

- (1) Only pair B – II is incorrect
- (2) Only pair C – III is incorrect
- (3) Both A – I and D – IV are incorrect
- (4) All pairs are correct

145. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Passageways up to the terminal bronchioles	(I)	Actual site of O_2/CO_2 diffusion across the respiratory membrane
(B)	The respiratory zone	(II)	The physical movement of air driven by pressure gradients
(C)	Pulmonary Ventilation	(III)	Filters, warms, and humidifies inspired atmospheric air
(D)	Mucus in respiratory tract	(IV)	A defense mechanism that traps and removes inhaled pathogens and debris.

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-II, D-IV
- (2) A-I, B-III, C-IV, D-II
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-I, C-IV, D-III

146. To initiate the high-pressure systemic circuit responsible for nourishing the body tissues, oxygenated blood is forcefully ejected from the **(I)** into the **(II)**.

- (1) **(I)** right ventricle, **(II)** pulmonary artery
- (2) **(I)** left ventricle, **(II)** aorta
- (3) **(I)** left atrium, **(II)** left ventricle
- (4) **(I)** right atrium, **(II)** vena cava

147. Read the following statements regarding the composition and function of lymph. Identify the **correct** statements and choose the appropriate option from below.

- A. Lymph is a coloured fluid because it contains the respiratory pigment haemoglobin, even though it lacks Red Blood Cells.
- B. The interstitial fluid, once it enters the network of lymphatic capillaries, is referred to as lymph.
- C. A key function of the lymphatic system is immunity, which is carried out by specialized lymphocytes present in the lymph.
- D. Lymph serves as an important medium for the transport of substances like hormones and nutrients.

- (1) A, B, and D
- (2) B and C only
- (3) B, C and D
- (4) A and C only

148. Given below are two statements.

Statement I: On holding breath, rising CO_2 and falling O_2 concentration are equally important changes in blood that would first lead to the urge in breath.

Statement II: The formation of oxyhaemoglobin in the lungs causes haemoglobin to act as a stronger acid, promoting the release of H^+ ions which facilitates the unloading of CO_2 from the blood.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

149. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The partial pressure of O_2 in the blood pumped by the left ventricle is slightly lower than the partial pressure of O_2 in the blood found in the pulmonary veins.

Reason R: A small fraction of deoxygenated blood from the bronchial circulation, which supplies the lung, mixes with the oxygenated blood returning to the left atrium.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

150. Read the following statements (A to D) and choose the option which states them as true (T) or False (F) **correctly**.

- A. The first heart sound, 'lub', corresponds to the closure of the atrioventricular (mitral and tricuspid) valves and marks the beginning of ventricular systole.
- B. The second heart sound, 'dub', is produced by the closure of the semilunar (aortic and pulmonary) valves at the onset of ventricular diastole.
- C. Chronic hypertension, defined by a blood pressure consistently above 140/90 mmHg, is a significant risk factor for damage to critical organs such as the brain and kidneys.
- D. An electrocardiogram (ECG) is an invasive procedure that directly measures the mechanical pumping action of the ventricles to assess heart health.

A B C D

- (1) F F F F
- (2) T T F F
- (3) F F F T
- (4) T T T F

151. Read the following statements about breathing and blood. How many of them are **correct**?

- A. Once a haemoglobin molecule is completely full of oxygen, it cannot carry any more oxygen.
- B. CO_2 from tissues actively moves into the blood and red blood cells, where it reacts with water to form carbonic acid.
- C. When haemoglobin that is carrying carbon dioxide (carbaminohaemoglobin) breaks apart, it releases haemoglobin and oxygen.
- D. In every 100 mL of deoxygenated blood, the plasma carries approximately 3.08 mL of CO_2 in various forms.
- E. C-shaped cartilage rings, which are open at the back, are found in the primary bronchi to prevent them from collapsing.

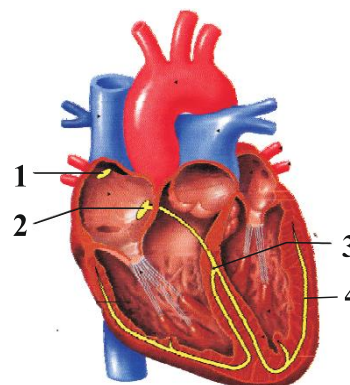
Choose the **correct** option:

- (1) One (2) Two
- (3) Three (4) Four

152. The sigmoid curve represents the oxygen-haemoglobin dissociation under normal metabolic conditions. During strenuous exercise, factors like increased pCO_2 , elevated temperature, and increased H^+ concentration are observed in the tissues. How would this condition alter the position of this curve and what is its physiological significance?

- (1) The curve would shift to the left, increasing haemoglobin's affinity for oxygen and promoting O_2 uptake.
- (2) The curve would not shift but oxygen release enhances.
- (3) The curve would shift to the right, decreasing haemoglobin's affinity for oxygen and facilitating O_2 unloading at the tissues.
- (4) The curve would shift to the right, increasing haemoglobin's affinity for oxygen and hindering O_2 release.

153. In the given section of the human heart which part ensures that ventricles do **not** contract prior to the atria?



- (1) 1 (2) 2
- (3) 3 (4) 4

154. Which of the following statements **accurately** describes the changes in the dimensions of the thoracic cavity during the process of inspiration?

- (1) Contraction of the external intercostal muscles pulls the ribs downward and inward, increasing the volume along the antero-posterior axis.
- (2) The diaphragm contracts and flattens, leading to an increase in the volume of the thoracic cavity in the antero-posterior axis.
- (3) The primary effect of the contraction of the external intercostal muscles is to lift the rib cage, which increases the thoracic volume in both the dorso-ventral axis and antero-posterior axis.
- (4) When the diaphragm contracts, it becomes more dome-shaped, thereby increasing the volume of the thoracic cavity along the dorso-ventral axis.

155. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The primary mode of oxygen transport from the alveoli to the body tissues is in the form of oxyhaemoglobin, not as dissolved gas.

Reason R: The solubility of oxygen in blood plasma is very low, making it an insufficient mechanism for meeting the metabolic demands of the body, whereas the oxygen-carrying capacity of whole blood is substantially higher.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true but R is not the correct explanation of A.
- (3) A is true but R is false.
- (4) A is false but R is true.

156. Match List-I with List-II.

List-I		List-II	
(A)	SA Node	(I)	Prevents the backflow of blood from the right ventricle into the right atrium.
(B)	Tricuspid Valve	(II)	Individual has A antigens on their RBCs and can safely donate to blood groups A and AB.
(C)	Blood Group O	(III)	Known as the "universal donor" as their RBCs lack A and B antigens.
(D)	Blood Group A	(IV)	Located in the upper right wall of the right atrium, it generates the primary impulse for a heartbeat.

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-III, D-II
- (2) A-IV, B-II, C-I, D-III
- (3) A-II, B-I, C-IV, D-III
- (4) A-I, B-IV, C-II, D-III

157. An individual's ECG shows that the R-R interval is consistently 1.0 second. What is their heart rate in beats per minute?

- (1) 72
- (2) 80
- (3) 60
- (4) 100

158. Which one of the following statements regarding the composition and function of human blood is **incorrect**?

- (1) Plasma contains a variety of ions, such as Na^+ , Ca^{2+} , and HCO_3^- , which are essential for various physiological processes.
- (2) The plasma protein albumin plays a primary role in maintaining the colloid osmotic pressure of the blood.
- (3) The globulin fraction of plasma proteins includes antibodies (immunoglobulins) that are fundamental to the body's immune defense.
- (4) To prevent spontaneous intravascular coagulation, platelets secrete anti coagulant.

159. Select the **incorrectly** matched pair with respect to the type of leucocyte and its normal relative percentage in human blood.

- (1) Neutrophils – 60-65%
- (2) Basophils – 16-18%
- (3) Monocytes – 6-8%
- (4) Lymphocytes – 20-25%

160. Given below are two statements.

Statement I: Under normal physiological conditions, an additional volume of air, an adult man can inspire by a forcible respiration averages 2100-2300.

Statement II: The diffusion membrane is made up of cuboidal epithelium of alveoli, the endothelium of alveoli capillaries and the basement substance in between them.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

161. If the cardiac output remains constant (5040 ml/minute) and the stroke volume increases to 84 ml, what would be the new heart rate?

- (1) 72 beats/minute
- (2) 60 beats/minute
- (3) 84 beats/minute
- (4) 50 beats/minute

162. Which of the following statements is/are **correct**?
- Normal composition of alveolar air is approximately 14% oxygen, 6% carbon dioxide and 80% nitrogen.
 - Adult frogs, when on land, rely exclusively on their lungs for gaseous exchange.
 - Reptilian lungs are more complex and possess a greater surface area for gas exchange compared to the simpler lungs of amphibians.
 - The tracheal system is the respiratory organ for insects, while book gills are used by most aquatic reptiles.
 - During inspiration lungs expand as the air rushes into the lungs.
- (1) A and C only (2) B and D only
(3) A, C, and E (4) B, D and E

163. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion A: While quiet expiration is a passive process that does not require muscular contraction, expiration can become an active, energy-expending process during activities like vigorous exercise or coughing.

Reason R: Passive expiration is driven by the elastic recoil of the lungs and chest wall upon contraction of the diaphragm, whereas active expiration is achieved by the contraction of abdominal muscles and internal intercostals to forcefully decrease thoracic volume.

In the light of the above statements, choose the **correct** answer from the options given below:

- A is true but R is false.
- A is false but R is true.
- Both A and R are true and R is the correct explanation of A.
- Both A and R are true but R is NOT the correct explanation of A.

164. A blood sample is prepared for pre-transfusion testing. Blood is tested with three different antisera with the following results:

- Test 1:** When mixed with Anti-A antiserum, significant agglutination (clumping) is observed.
- Test 2:** When mixed with Anti-B antiserum, no agglutination is seen.
- Test 3:** When mixed with Anti-Rh antiserum, no agglutination is seen.

Based on these results, what is the **correct** blood type of the sample?

- A positive (A^+)
- A negative (A^-)
- AB negative (AB^-)
- O negative (O^-)

165. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Larynx	(I)	Cartilaginous sound box
(B)	Epiglottis	(II)	Enzyme mainly present in RBCs
(C)	Carbonic anhydrase	(III)	Prevents entry of food in wind pipe
(D)	CO ₂ transport by RBC	(IV)	20-25%

Choose the **correct** answer from the options given below:

- A-I, B-III, C-II, D-IV
- A-IV, B-I, C-III, D-II
- A-I, B-II, C-III, D-IV
- A-I, B-II, C-IV, D-III

166. In which of the following organisms is a closed circulatory system found?

- Arthropods and Molluscs
- Annelids and Chordates
- Sponges and Coelenterates
- Molluscs and Annelids

167. Given below are two statements.

Statement I: The condition named *erythroblastosis foetalis* can be avoided by administering anti-Rh antibodies to the mother immediately before the delivery of the first child.

Statement II: Chordae tendinae, are thread-like structures attached to the flaps of bicuspid and tricuspid valves at one end and their other ends are attached to auricular wall.

In the light of the above statements, choose the most appropriate answer from the options given below:

- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.
- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.

168. Carefully read the following statements and select the **correct** option:

- The normal breathing rate of a human is 12-16 times per minute.
- Respiration is most efficient in insects among invertebrates.
- Inhaled air has 21% oxygen whereas exhaled oxygen has 16% oxygen.
- Arteries are deep-seated in the body, whereas veins are more superficial.
- Birds have a 4-chambered heart and exhibit double circulation.

- Only three statements are correct
- Only four statements are correct
- All five statements are correct
- Only two statements are correct

169. The evolution of a closed circulatory system was a critical adaptation for supporting a more active lifestyle and larger body size compared to animals with open systems, like arthropods. What is the primary reason for this?

- (1) A closed system allows the circulatory fluid (blood) to mix directly with the interstitial fluid, which enhances nutrient delivery.
- (2) In a closed system, the confinement of blood within vessels allows for the maintenance of high pressure and rapid flow, enabling efficient and targeted delivery of oxygen and nutrients.
- (3) An open system operates at a much higher pressure, requiring a more robust and energetically expensive heart.
- (4) The total volume of circulatory fluid (hemolymph) in an open system is significantly smaller than the blood volume in a closed system of a similar-sized animal.

170. To measure a person's expiratory reserve volume, you would ask them to:

- (1) inhale normally and then exhale as much as possible.
- (2) exhale normally and then exhale as much as possible.
- (3) inhale as much as possible and then exhale normally.
- (4) breathe normally for several cycles.

171. Which granulocytes are primarily responsible for secreting histamine and serotonin to promote vasodilation?

- (1) Neutrophils
- (2) Eosinophils
- (3) Basophils
- (4) Monocytes

172. Read the following statements concerning the roles of various blood cells and choose the option with the **correct** combination of statements.

- A. Mast cells are responsible for initiating inflammatory responses.
- B. The phagocytosis of pathogens like bacteria is a primary function carried out by neutrophils and monocytes.
- C. The specific, long-term immunity of the body is primarily dependent on the function of granulocytes like basophils and eosinophils.
- D. Oxygen transport to the tissues is a function shared by both erythrocytes and basophils.

Which of the following options contains only the **correct** statements?

- (1) A and B only
- (2) C and D only
- (3) A, B and C
- (4) B and D only

173. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Total lung capacity	(I)	Helps workers working in stone grinding factories
(B)	Mask	(II)	5800 ml
(C)	Cardiac arrest	(III)	Sudden cessation of heart beat
(D)	Coronary system	(IV)	Provides nutrients and oxygen to the cardiac musculature

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-IV, C-I, D-III

174. Given below are two statements.

Statement I: In a healthy person, the amount of air that is always left in the lungs even after you force out all you can is greater than the extra amount of air you can force out after a normal exhale.

Statement II: In a person with lung diseases like emphysema or fibrosis, the Vital Capacity is smaller than normal, but the amount of air they breathe in and out during quiet, regular breathing can still be normal.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

175. Which of the following **best** describes the main difference between a heart attack and heart failure?

- (1) A heart attack is the death of heart muscle from a blocked blood supply, whereas heart failure is the heart's inability to pump effectively.
- (2) Heart failure is a sudden event where the heart stops beating, while a heart attack is a chronic, long-term condition.
- (3) A heart attack is a problem with the heart's overall pumping efficiency, while heart failure is caused by a sudden artery blockage.
- (4) Heart failure and heart attack are two different names for the exact same medical problem.

176. Read the following statements and determine how many of them are **correct**.

- A. Due to its higher solubility, the diffusion rate of carbon dioxide across the respiratory membrane is approximately 20 times faster than that of oxygen.
- B. The deoxygenated blood is significantly saturated with oxygen and is not completely devoid of it.
- C. Each gram (g) of haemoglobin, when fully saturated, can transport a maximum of about 1.34 mL of oxygen.
- D. The intercostal muscles, which are crucial for changing the volume of the thoracic cavity, are located in the spaces between adjacent ribs.
- E. Lymphocytes are of two major types – ‘B’ and ‘T’ forms.

How many of the above statements are **correct**?

- (1) Two
- (2) Three
- (3) Four
- (4) Five

177. Which of the following statements about an electrocardiogram (ECG) is **true**?

- (1) The P-wave represents the electrical signal for the contraction of the heart's lower chambers (ventricles).
- (2) The QRS complex shows the powerful electrical signal that makes the ventricles contract and pump blood.
- (3) The T-wave represents the electrical activity when the heart's upper chambers (atria) contract.
- (4) A person's heart rate is calculated by measuring the time from the start of the QRS complex to the end of the T-wave.



178. If a splenectomy (surgical removal of the spleen) is performed on a mammal, which of the following is the **most probable** long-term consequence?

- (1) A decrease in the rate of erythropoiesis in the red bone marrow.
- (2) The animal would lose its primary site for the production of erythrocytes.
- (3) A noticeable increase in the population of circulating old age and damaged erythrocytes.
- (4) The average lifespan of a circulating erythrocyte would extend well beyond 120 days.

179. Select the **incorrect** match.

- (1) Bundle of His– Located in the auriculo-ventricular septum
- (2) Papillary muscles– Prevent the cusps of AV valves from everting into the atria during ventricular Contraction
- (3) 100 ml oxygenated blood – Delivers 5 mL of oxygen to tissues under normal physiological condition
- (4) Each haemoglobin – Can carry maximum four molecules at a time

180. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The combination of oxygen with the iron atom of haemoglobin is a physical process of oxygenation, not a chemical process of oxidation.

Reason R: The binding of oxygen to the haemoglobin molecule is readily reversible, and the iron atom remains in its ferrous (Fe^{2+}) state, which is essential for it to release oxygen to the tissues.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

SPACE FOR ROUGH WORK



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>

Test Analysis Sheet

- Student Name :
- PW Registration No. :
- Batch Code :
- Test Name/Type :

Max. Marks:
Marks Obtained:
Percentage:
Date of Exam :

1. Subject-wise Performance

Subject	Attempt Order	Time taken (min)	Marks Expected	No. of Attempted Q's	No. of Wrong Q's	Accuracy (%)
Physics						
Chemistry						
Botany						
Zoology						

- Was there a rush at the end? (Yes/No): _____
- Did you revise after solving? (Yes/No): _____

2. Wrong Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Calculation mistake					
Silly mistake					
Marked incorrect option, however solved correctly					
Conceptual Error					
Others					

3. Unattempted Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Forgot the concept / Formula					
Solved correctly but didn't have confidence to mark it on OMR					
Missed the class /Didn't revise					
Got confused with the concept					
Others					

4. Weaknesses Identified

(List topics/chapters where you lost marks)

- Physics :
- Chemistry :
- Botany :
- Zoology :

5. Action Plan Before Next Test

1. Types of Mistakes to Avoid:
2. Extra Practice Needed in:
3. Strategy Changes: (e.g., attempt Chemistry first, manage time better):
4. What would you like to improve emotionally/mentally for the next test? :

Student Signature: _____